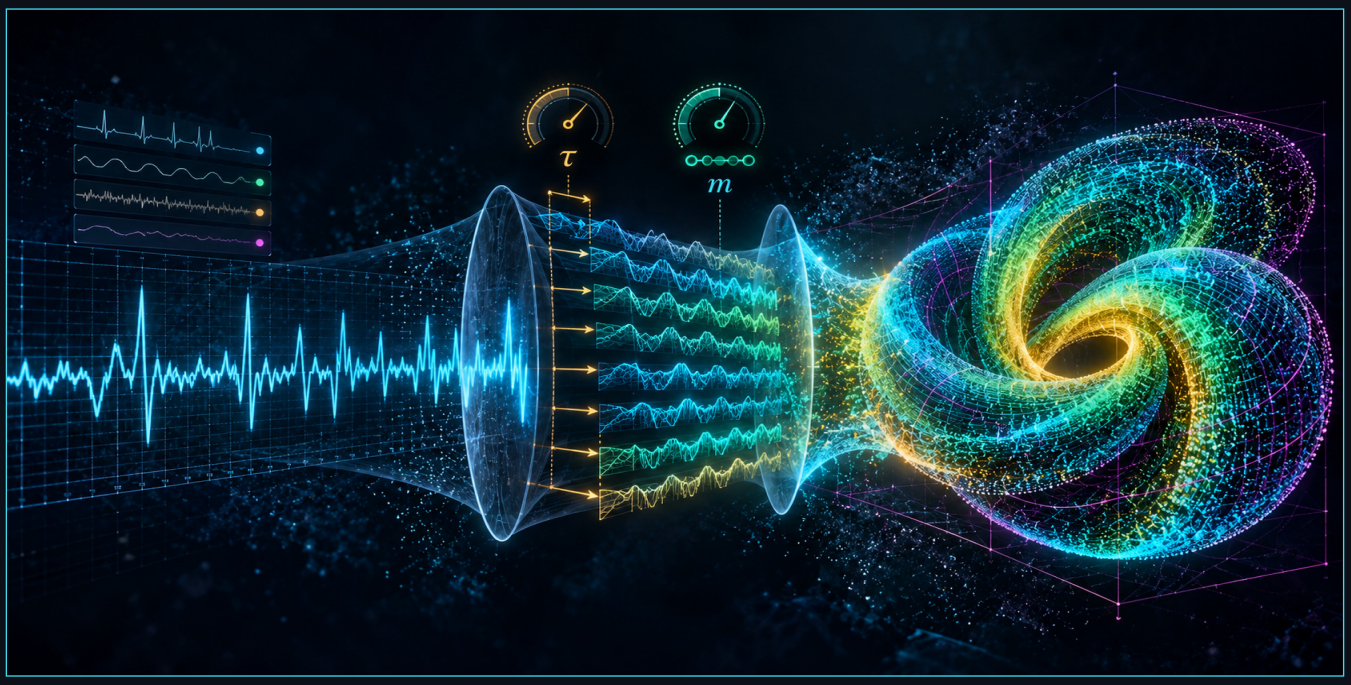
TRAJECTORY MEDICINE FRAMEWORK

A process-centred research architecture
for longitudinal physiological signal analysis



BIOS > INFO > META / HUMAN REVIEW

RESEARCH FRAMEWORK / NO CLINICAL CLAIMS

Krzysztof Baran · LifeNode Research Collective

Expanded illustrated edition v1.1 · 31 October 2026

PROCESS > STATE

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LIFENODE / TRAJECTORY MEDICINE / 00

Editorial protocol

An illustrated field atlas for a falsifiable research programme - not a declaration of clinical success.

This expanded edition reorganises the Trajectory Medicine Framework (TMF) into a research narrative with explicit decision gates, measurement contracts and visual research plates. It preserves the intellectual ambition of the LifeNode programme while separating established mathematical tools, observed biological phenomena, engineering proposals and untested medical hypotheses.

The visual language follows the LifeNode Guerilla DeSci / Tokio Drift manifesto: near-black architecture, cyan analytical lines, magenta failure markers and amber provenance seals. That is an editorial language, not a change in the evidentiary standard. An image of a mitochondrial tether, attractor, bio-hybrid chamber or validation monument is an explanatory model; it is not a photograph of a completed LifeNode experiment.

SOURCE AND SCOPE
Built from LifeNode TMF whitepaper v1.0 (31 October 2026), six supplied research illustrations, and the LifeNode Guerilla DeSci manifesto as a graphic reference. Additions in this edition are proposed research-design refinements; no new experimental results or independent validation are claimed.

Reader navigation

ACT	QUESTION	WHERE TO LOOK
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III · TEST	How could the central claim be disproved?	Sections 9–12 · Plate 04
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REPRODUCE	What must an independent laboratory receive?	Appendices A–C · Source register

Evidence legend

TAG	MEANING
OBSERVATION	Measured result tied to an identifiable external study or LifeNode record.

TAG	MEANING
FORMALISM	Established mathematical structure whose assumptions must be checked.
CONSTRUCTION	Project-defined metric, instrument or operational architecture.
HYPOTHESIS	Specific, testable claim not yet validated.
INTERPRETATION	Proposed explanation that is weaker than direct proof.
VALIDATED	Result satisfying a frozen protocol, adequate baselines, replication and provenance.

LIFENODE / TRAJECTORY MEDICINE / 01

Executive brief

One central research question; six falsification gates; no shortcut from retrospective signal analysis to patient-facing medicine.

TMF asks whether the evolution of a physiological signal carries reliable information beyond conventional summaries and existing signal-processing methods. Candidate information includes recurrence, rate and direction of change, coupling and delay across channels, and the geometry of reconstructed trajectories. It is a claim about possible **incremental information**, not a claim that established clinical measurements are obsolete.

The proper comparison is not trajectory analysis versus an inert snapshot. Clinical medicine already employs ambulatory ECG, telemetry, EEG, HRV, blood pressure time series, longitudinal laboratory data and risk models. TMF is justified only when its added descriptors outperform strong, appropriately tuned alternatives on properly separated data.

The immediate research implementation is offline, retrospective and hardware-independent. It should first establish a narrow ECG rhythm-characterisation benchmark, using reproducible signal handling and held-out subjects. That initial exercise cannot validate a 24–48-hour warning horizon. A later forecasting study would need long enough pre-event recordings, reliable event times and explicit censoring rules.

> DECISION THESIS
> A compelling trajectory visualisation is not a clinically meaningful predictor. The essential experiment is whether the added trajectory features survive strong baselines, leakage controls, structured nulls, an independent dataset and prospective silent evaluation.

Four commitments that constrain the programme

- Preserve raw signal and acquisition provenance; never confuse reconstructed geometry with direct access to the full biological state.
- Freeze the exact ASCALON metric, units and normalisation before analysing held-out test data.

- Treat HMF bio-hybrid material research as an independent engineering branch; its bench performance cannot authenticate a clinical claim.
- Publish failures with the same reproducibility and DOI discipline as passes, including the code and the test that failed.

LIFENODE / TRAJECTORY MEDICINE / 02

Boundary of the claim

Define intended use before optimising an instrument or a mathematical score.

2.1 Current intended use

The current system target is a non-actionable research pipeline for de-identified physiological time series. It may emit signal-quality flags, reconstructed coordinates, descriptive features, uncertainties and model-comparison reports. Its output is not an individual diagnosis, an emergency indication or a treatment recommendation.

An observed trajectory displacement can be useful even if it is caused by activity, contact pressure, motion or sleep. In that case it is an acquisition or context feature, not evidence of disease. Endpoint adjudication and covariate analysis decide which interpretation remains tenable.

2.2 Explicit exclusions

- No validated arrhythmia, seizure, infection, psychiatric or heart-failure prediction system exists on the basis of this paper.
- No confirmed 24–48-hour lead time or universal clinical threshold for θ is demonstrated.
- No patient-facing alert, autonomous triage, feedback stimulation or closed-loop therapy is part of this research release.
- No claim that analogue-to-digital conversion inherently destroys physiological timing or phase information.
- No inference that mitochondrial morphology, fungal spikes or attractive mathematical shapes independently validate a human biomarker.

2.3 Claim-to-evidence ladder

RESEARCH OUTPUT	NOW SUPPORTS	DOES NOT YET SUPPORT
Delay embedding	Representation of recorded samples given a specified τ, m and preprocessing chain.	Proof of a biological attractor or a symplectic manifold.
ASCALON candidate	A versioned exploratory statistic if its definition and estimator are reproducible.	Universal health score or fixed medical cut-off.
Retrospective classifier	Association on the held-out records and endpoint defined by that study.	Prospective clinical utility or a future-event warning.
HMF bench measurement	Transfer, impedance, phase and drift properties under test conditions.	A safe human sensor, much less a disease detector.

> EPISTEMIC FIREWALL
> A research signal may be interesting, correctly computed and externally reproducible while still being clinically irrelevant. Each transition across this ladder requires separate evidence.

LIFENODE / TRAJECTORY MEDICINE / 03

Why biological geometry matters

Specific structure–function examples motivate measurement; they do not validate an entire medical ontology.

3.1 Mitochondrial spacing as a functional variable

The 2026 SPRIZ / CCDC127 study [1] reports an approximately 18-nm tether spanning the mitochondrial intermembrane space. In experimental systems, changing this protein altered local spacing and was associated with measurable changes in accessibility and mitochondrial Ca^{2+} uptake. The observation is mechanistically specific: organelle geometry can be regulated and can matter to function.

The transfer to TMF is methodological, not evidentiary. Biological effects may depend on spatial relationships that summary scalar measurements fail to describe completely. But the existence of a mitochondrial tether says nothing directly about θ, ECG risk horizons or a universal topological pathology signature. A scale bridge would need its own observable, mechanism and experiment.

3.2 Mycelial electrical propagation as a time-scale lesson

Adamatzky [2] reported slow, millivolt-scale electrical events in oyster-fungus-colonised substrates, with ordered lead–lag relationships across an eight-channel linear electrode array. The cited study sampled at 1 Hz over roughly fifteen days and used shuffled controls to probe the directional structure. Its time scale and geometry should not be imported into human cardiac electrophysiology.

The methodological opportunity is clear: synchronised multi-channel acquisition allows directional hypotheses that a single aggregate number cannot resolve. The equipment also demonstrates that a properly engineered ADC acquisition chain is fully compatible with research into slow biological dynamics.

> INFERENCE LIMIT
> Structure–function coupling and distributed electrophysiology are observed in their own domains. TMF is a separate programme that must prove any useful human-signal feature empirically.

RESEARCH PLATE 01 / MITO

Mitochondrial architecture / a functional boundary



FIGURE 1. Conceptual reconstruction of the mitochondrial intermembrane region, with repeated tether-like structures and enlarged interface details.

HOW TO READ THIS PLATE

Read the repeating bridges as a visual encoding of local spacing constraints. The relevant research proposition is that geometry can affect function; the drawing does not identify individual molecules experimentally or measure a patient signal.

EPISTEMIC STATUS External biological motivation [1]; interpretive illustration, not an imaging result from LifeNode.

REPRESENTS	REQUIRES A TEST	DOES NOT ESTABLISH
Cell architecture / physical spacing	Compare morphological conditions and functional consequences	Proof of a universal human diagnostic geometry

LIFENODE / TRAJECTORY MEDICINE / 04

System architecture: BIOS → INFO → META

Keep biology, measurement and interpretation separable, auditable and incapable of silently becoming a treatment loop.

4.1 BIOS / substrate and instrumentation

BIOS is the biological source together with the conditions under which a measurable signal exists. It includes organism, sampling site, sensor, electrode configuration, environmental context and acquisition chain. Candidate human modalities are ECG, PPG, respiration, EEG, blood pressure, temperature and accelerometry; mycelial signals remain in a different research domain.

The instrument cannot be treated as a transparent window. Coupling gel, contact impedance, analogue filters, firmware, clocks and missing-data mechanisms are potential determinants of the apparent trajectory. Instrument state is therefore part of the evidence record, not an implementation footnote.

4.2 INFO / deterministic analytics

INFO handles quality control, temporal alignment, preprocessing, embedding, feature extraction, standard baselines, structured nulls and statistical evaluation. Every transformation should be reproducible. Randomised algorithms require seeds, versioned dependencies and complete parameter files; operations tuned on data must be fitted only within the training partition.

TMF must store conventional baseline features and trajectory candidates side by side. Without a direct comparison on identical samples, a sophisticated new feature family can only appear superior because of dataset selection, model capacity or unequal tuning.

4.3 META / interpretation and human review

META records the question that motivated the analysis, dataset licence, intended population, endpoint, model uncertainty and decision history. Human review provides meaning to a research outcome but does not convert model output into a medical recommendation. Clinical action remains absent from the current system architecture.

The Loop I / Loop II distinction is useful here: the diagnostic loop can observe and characterise; a living feedback or coupling loop would be a different, more hazardous engineering and regulatory undertaking. The present TMF whitepaper does not authorise that second loop.

4.4 Interface contract

BOUNDARY	REQUIRED INPUT	AUDITABLE OUTPUT
BIOS → acquisition	Physical signal, clock, sensor settings, context.	Raw files, timebase, calibrations, missingness and quality flags.
Acquisition → INFO	Frozen manifest, channel map, no hidden preprocessing.	Features, hashes, software versions, quality and null reports.

BOUNDARY	REQUIRED INPUT	AUDITABLE OUTPUT
INFO → META	Model comparison, uncertainty, subgroup analysis.	Written claim, limitations, pass/fail/invalid decision, reviewer identity.

Acquisition is an information contract

Preserve the temporal structure actually measurable by the sensor; do not mystify analogue or digital signals.

A correctly designed analogue front-end, anti-alias filter and ADC can retain the information required by a chosen analysis. The failure modes are specific: insufficient effective bandwidth, saturation, jitter, clock drift, quantisation noise, unstable electrode potential, undocumented filtering and motion contamination. “No ADC” is not a valid replacement for quantitative transfer-function testing.

Acquisition requirements must follow the phenomenon. Millisecond ECG morphology cannot be treated like minute-scale mycelial spikes. Cross-channel synchronisation and precise event timestamps become decisive when the hypothesis concerns propagation, delay, coupling or an early-warning lead time.

5.1 Modality-aware design

MODALITY	CRITICAL ACQUISITION DETAIL	PRIMARY CONFUSION RISK
ECG	Clinical-grade bandwidth; lead configuration; accurate timebase; beat annotations.	Signal-quality or preprocessing artefact mistaken for new geometry.
PPG	Pulse morphology, contact quality, motion, optical channel metadata.	Perfusion and motion mistaken for path instability.
EEG	Montage, reference, impedance, aligned event markers and stage/context labels.	Sleep-stage and artefact effects mistaken for a disease precursor.
Respiration	Sensor calibration, posture and synchronisation with other channels.	Behavioural pacing mistaken for physiological transition.
Mycelium	Electrode arrangement, ionic environment, hydration, temperature, long-term drift.	Electrode polarisation mistaken for biological propagation.

5.2 Minimum raw-data manifest

- Device, firmware, gain, dynamic range, ADC resolution, analogue passband and anti-alias characteristics.
- Nominal/effective sampling rate; absolute time source; synchronisation offsets; observed clock drift.
- Lead and electrode map, contact material, sensor position, calibration and quality assessment.
- Missing periods, saturation events, data corruption, acquisition pauses and any hidden vendor processing.

- Raw data checksum and full derivation chain to each analysis-ready representation.

LIFENODE / TRAJECTORY MEDICINE / 06

Mathematical reconstruction / what the geometry does

A state-space representation is a model built from samples, not an automatic revelation of the underlying organism.

6.1 Delay coordinates

For an observed scalar process x(t), a conventional reconstruction is **y(t) = [x(t), x(t−τ), …, x(t−(m−1)τ)]**. Here τ is the lag and m is the embedding dimension. Takens-style reconstruction has formal guarantees only under specified dynamical and observability assumptions [5–7]. Human physiology typically includes noise, changing regimes, unobserved inputs and interventions.

The testable engineering task is to estimate τ and m on training data, then assess sensitivity to the chosen window, filtering, resampling and subject population. Parameter instability is a result to record; it is not solved by assigning an aesthetically convenient torus.

6.2 Direction-dependent metrics

Finsler geometry admits a local length function F(x,v) dependent on position and direction, subject to requirements including regularity, positivity and homogeneity. If TMF uses this formalism, it must identify the actual function, coordinate units, scaling conventions, domain and numerical estimator. A notation such as $g_i(x,v)$ without a defined F is not yet a metric implementation.

An even-dimensional reconstructed coordinate list is not sufficient to claim symplectic structure. That claim requires a specified closed, non-degenerate two-form and justified coordinates; physiological data may instead come from driven dissipative dynamics.

6.3 NLSE, Floquet and topology

NLSE dynamics, Floquet analysis, persistent homology and related mathematical objects can motivate candidate descriptors. They do not become biological mechanisms by analogy. For each imported tool, TMF must state the measurable observable, identifiability conditions, predicted advantage, suitable null and falsifier. More elaborate mathematics increases documentation obligations, not the default evidence level.

> RECONSTRUCTION CRITERION
> A trajectory feature remains interesting only if its computed variation can be distinguished from timebase, amplitude, measurement chain, preprocessing and generic autocorrelation effects.

RESEARCH PLATE 02 / TRAJECTORY

From signal to reconstructed trajectory

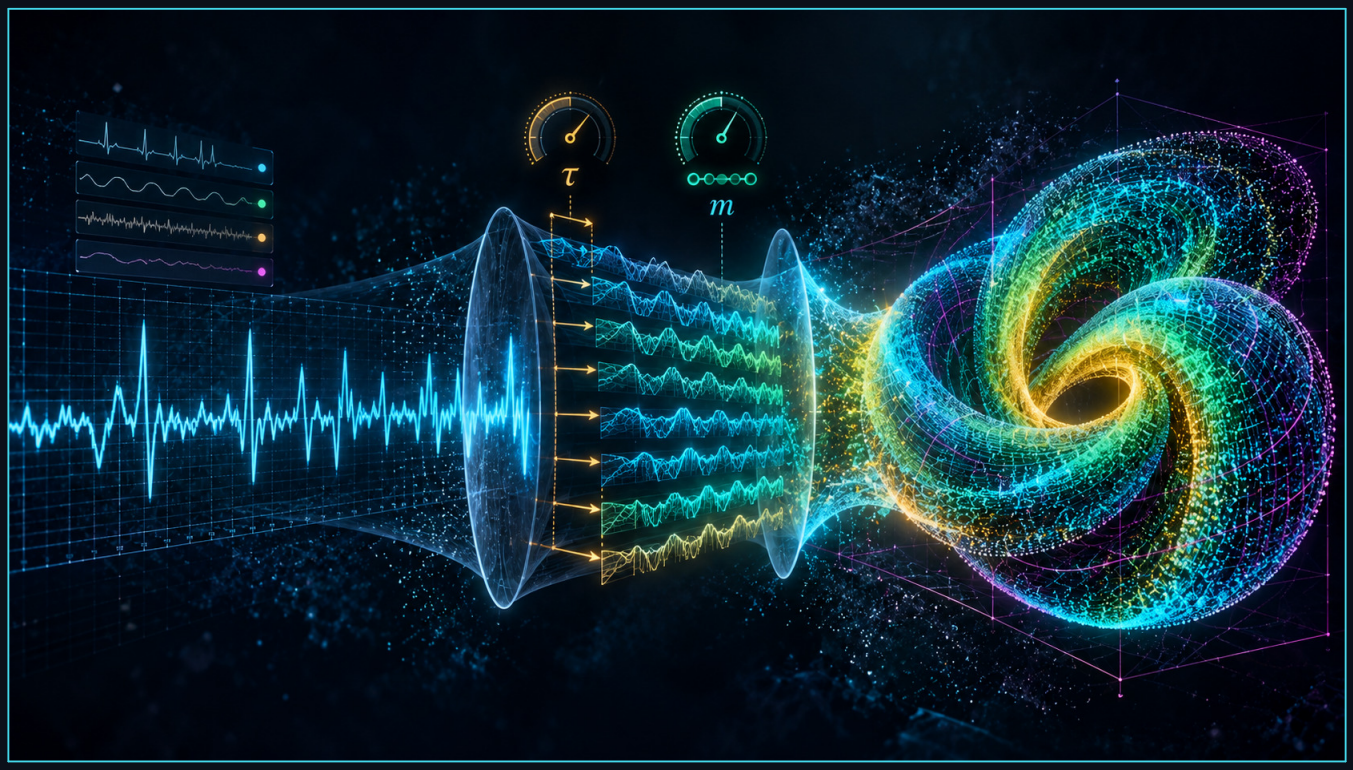


FIGURE 2. Conceptual flow from multichannel physiological time series through delayed coordinates and embedding parameters τ and m to a three-dimensional trajectory representation.

HOW TO READ THIS PLATE

The toroidal motif is a visual metaphor for one possible reconstructed structure, not evidence that living systems possess a universal toroidal attractor. The decisive evidence lies in parameter stability, null-control separation and incremental held-out performance.

EPISTEMIC STATUS Established delay-coordinate formalism [5–7] plus LifeNode interpretive construction; no human-biomarker validation.

REPRESENTS	REQUIRES A TEST	DOES NOT ESTABLISH
Recorded samples / chosen embedding	Lag, dimension, surrogates, baselines, perturbations	A confirmed toroidal biological attractor

LIFENODE / TRAJECTORY MEDICINE / 07

ASCALON / a metric that must earn its units

Before a clinical question can be asked, the mathematical object itself must be frozen and tested.

In the LifeNode source corpus, ASCALON θ appears as a ratio involving acceleration magnitude, trajectory speed and a kinetic-energy-like denominator. One stated form is $\theta_{raw} = [\int ||y''(t)||\,||y'(t)||\,dt] / [\int ||y'(t)||^2\,dt]$. A second description invokes a weighting function σ(s). These alternatives cannot silently be treated as the same operational definition.

Acceleration magnitude is not equivalent to geometric curvature. For a regular Euclidean curve, curvature depends on the component of acceleration orthogonal to velocity: $\kappa = ||v \times a|| / ||v||^3$. Finsler geodesic curvature would additionally require a defined connection. Naming the numerator “curvature” without this work introduces an avoidable category error.

With physical time in the displayed ratio, the units are inverse time. An unspecified mapping into [0,1] cannot guarantee portability between sampling conventions or observation horizons. Consequently, θ = 0.70 is a project hypothesis, not a medical cut-off, and must not be used for patient decisions.

7.1 Freeze a normative specification

- One continuous equation and one explicitly derived discrete estimator, each assigned a version identifier.
- All units, integration conventions, derivative approximations, regularisation and near-zero-velocity guards.
- A deterministic normalisation pipeline with training-only reference distribution and defined handling of unseen subjects.
- Predeclared sensitivity tests for sampling rate, time dilation, amplitude scaling, coordinate changes, noise and missingness.
- The exact allowable parameter search and fitting method for τ, m, window duration and smoothing strength.
- A calibration and uncertainty strategy distinct from the feature computation itself.

7.2 The minimum negative-control battery

Test the estimator on constant signals, linear ramps, known smooth curves, sinusoidal mixtures, noisy stochastic processes and time-warped synthetic trajectories. Compare independent implementations. Document when the numerator becomes unstable or the denominator approaches zero. A feature that fails elementary known-answer tests cannot advance to a medical dataset, regardless of a high downstream AUC.

> METRIC STATUS / NO THRESHOLD
> Until the estimator, units, normalisation and acquisition sensitivity have passed F0 and F1, θ must be reported as an exploratory research construction only.

LIFENODE / TRAJECTORY MEDICINE / 08

HMF / independent materials-and-transduction branch

A bio-hybrid interface is a measurable material system, not a validated phase lens or a patient sensor.

HMF proposes hydrogel-based ionic-to-electronic transduction in controlled laboratory experiments, with alginate/chitosan/PVA variants, optional conductive polymer components, electrode arrays and environmental controls. It can be studied without assuming clinical significance or coupling to human tissues.

The core measurement is an input/output instrument response. Does the gel amplify, attenuate, delay, distort or drift relative to a calibrated reference? Does geometry change observability independently of electrode surface area and contact quality? Can the measured activity be distinguished from electrochemistry, hydration transients and ambient noise?

8.1 Experimental matrix

QUESTION	MEASUREMENT	DEFEATING CONTROL
Signal modification	Transfer function, gain, phase error, impedance and dynamic range.	Calibrated reference input; dry/hydrated/colonised conditions.
Stability	Hours-to-days drift, hysteresis, hydration, temperature and ageing.	Controlled humidity, ionic strength and thermal cycling.
Biological origin	Replicate events with substrate and environment metadata.	Abiotic blank, uninoculated substrate, disconnected channels.
Electrode layout	Matched linear, star/interdigitated and planar geometries.	Equalise active area, distance, materials and sampling chain.
Information preservation	Reference waveform reconstruction and error budget.	Independent recorder and known-answer artificial signals.

8.2 The separation rule

A successful HMF bench run would establish a specified transduction property under controlled conditions. It would not validate ECG trajectory scoring or show that a hydrogel membrane is safe for prolonged human wear. Clinical interface development would require a new intended-use statement, biocompatibility and risk work, and new evidence.

RESEARCH PLATE 03 / HMF

HMF bench architecture / four measurements

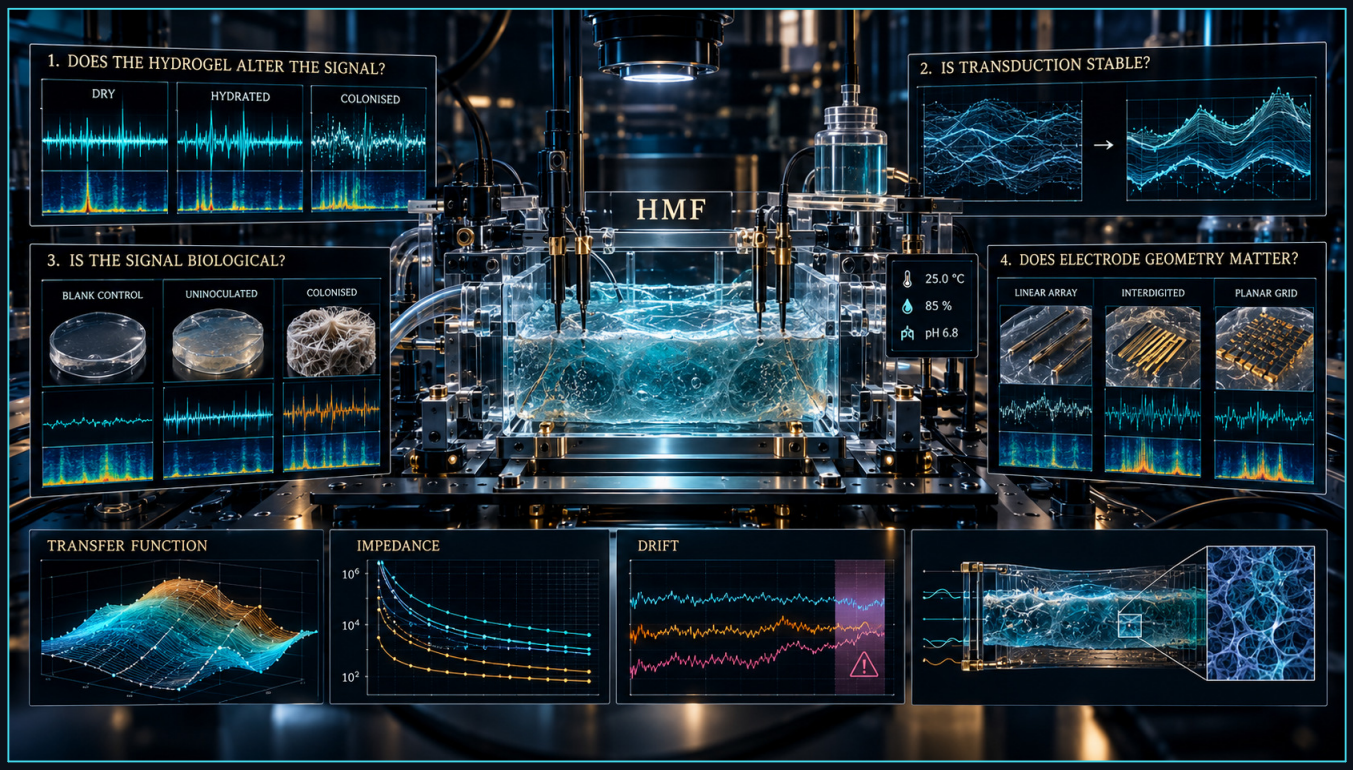


FIGURE 3. Proposed research chamber with comparative conditions, electrical readouts, geometry options and drift/impedance characterisation.

HOW TO READ THIS PLATE

The four questions - whether the medium changes the signal, remains stable, contains a biological component and depends on electrode geometry - are experimental obligations. The depicted screens illustrate required data products, not results collected by LifeNode.

EPISTEMIC STATUS LifeNode engineering proposal [10,11]; material properties and phase-response claims require bench measurements.

REPRESENTS	REQUIRES A TEST	DOES NOT ESTABLISH
Bench inputs / reference electrodes	Impedance, phase response, stability, blank controls	A validated wearable or biological phase lens

LIFENODE / TRAJECTORY MEDICINE / 09

Zero-Build / the smallest credible experiment

A complete negative result on one well-specified endpoint is more valuable than an unconstrained multi-disease model.

9.1 The first endpoint

The recommended first software benchmark is an annotated ECG rhythm-characterisation task over fixed observation windows, with a held-out subject design. The MIT-BIH Arrhythmia Database is useful for testing loaders, beat annotations, waveform/HRV features and subject-aware evaluation. It is not a 24–48-hour pre-event forecasting dataset by virtue of containing ECG and event labels.

For a future-warning claim, define the clinical event, lead-time horizon, observation cutoff, censoring, competing events and alert aggregation. The dataset must contain enough pre-event time and valid endpoint dates. Back-projecting a short rhythm-classification task into a long warning horizon would be an endpoint substitution.

9.2 Two comparisons, not one

The comparison must cover **feature sufficiency** (can conventional data summaries already explain the endpoint?) and **predictive increment** (does adding frozen trajectory features produce a meaningful, replicable gain?). Fit the same model family and input windows where possible. Report tuning budgets, split IDs, confidence intervals and paired uncertainty for the performance difference.

Separate the potential added value of generic nonlinear time-series features from the project-specific contribution of ASCALON. A performance improvement due solely to well-known recurrence or entropy measures does not yet validate a novel θ construction.

9.3 Benchmark gates

STAGE	OPERATIVE QUESTION	REQUIRED ARTIFACT
Z0 / integrity	Are records, labels, licences and subject splits reconstructible?	Manifest, data audit and frozen cohort specification.
Z1 / baseline	How much do standard features and supervised classifiers explain?	Strong baseline report and calibration analysis.
Z2 / increment	Does frozen trajectory geometry add held-out value?	Ablation report and paired comparison.
Z3 / robustness	Do nulls, noise and leakage checks destroy a false gain?	Null distributions and sensitivity report.
Z4 / external	Does a locked specification transport to a new cohort?	Replication or explicit non-replication report.

STAGE	OPERATIVE QUESTION	REQUIRED ARTIFACT
Z5 / silent use	Would this remain useful without acting on patients?	Prospective shadow-mode evaluation plan and outcomes.

Leakage, baselines and meaningful metrics

A good-looking classification curve can still be a broken experiment.

10.1 Partition discipline

- Split at subject level: overlapping windows from one person cannot appear in both train and test.
- Keep all windows belonging to one clinical episode within a single partition.
- Fit scaling, lag, embedding dimension, feature selection, calibration and threshold on training data only.
- If the claim is future prediction, employ temporal cutoffs that prohibit post-event information.
- Treat dataset, device and acquisition site as possible confounders and document subgroup failures.

10.2 Metrics matched to the use case

For rhythm classification, report AUROC, AUPRC, sensitivity, specificity, positive and negative predictive value, calibration and uncertainty intervals; aggregate properly at the subject level. For event warning, report event sensitivity, false alerts per subject-day, alert-time burden, lead-time distributions and calibration at the intended horizon.

A significant p-value does not imply useful alert performance. A retrospective score may achieve attractive discrimination while producing more false notifications than a real clinical workflow can tolerate. Decision-curve and net-benefit analyses become relevant only when the target intervention and harm trade-off have been specified.

10.3 What counts as incremental value

Declare a clinically or operationally meaningful minimum difference **before** opening the test set. Compare the TMF extension to the strongest justified conventional model with matched information access. Include a feature-family ablation: conventional only; conventional plus standard nonlinear; conventional plus ASCALON; conventional plus all TMF candidates. The outcome can be PASS, FAIL or INCONCLUSIVE with uncertainty and limitations.

LIFENODE / TRAJECTORY MEDICINE / 11

Falsification as the operating system

Passing a test opens no automatic door to the next claim.

TMF should advance through six explicit gates, F0–F5. Each gate tests a different kind of claim. Mathematical consistency is not signal robustness; robustness is not incremental prediction; prediction is not transportability; transportability is not clinical utility.

Negative results must be frozen alongside successful ones. A failed threshold is not edited retroactively until it passes; an amendment creates a new version, explains the motivation and requires a new independent test. This is also the role of LifeNode's append-only provenance and epistemic relay discipline.

11.1 Gate acceptance criteria

GATE	PASS REQUIRES	FAILURE MEANS
F0 / correct	Independent agreement and synthetic known-answer checks.	Estimator or specification not reliable.
F1 / robust	Predeclared tolerance under benign perturbations and preprocessing.	Possible measurement artefact.
F2 / value	Held-out increment over strongest baseline with uncertainty.	No demonstrated added value.
F3 / null	Performance degrades under appropriate surrogates and label controls.	May capture leakage or generic structure.
F4 / replicate	Locked code/threshold works on independent data.	Not yet transportable.
F5 / shadow	Silent prospective evaluation meets predeclared event/alarm constraints.	Not yet suitable for clinical workflow.

NEGATIVE RESULTS ARE PUBLICATION-GRADE

Record the exact failing gate, dataset, metric version, split, code hash, parameter bundle, uncertainty and next test. Give failure and success the same archival standard; do not retune the final test retrospectively.

RESEARCH PLATE 04 / GATES

F0–F5 / evidence gates and the negative result



FIGURE 4. Conceptual falsification ladder rising from metric correctness to prospective shadow validation, paired with an equally prominent archive for negative results.

> HOW TO READ THIS PLATE
> The locks are deliberate: satisfying F0 gives no authority to claim F1–F5. The rejected threshold-adjustment mechanism represents a versioning rule, not the assertion that a particular test already failed.

EPISTEMIC STATUS Proposed validation governance; the depicted pass/fail indicators are illustrative, not a LifeNode clinical result.

REPRESENTS	REQUIRES A TEST	DOES NOT ESTABLISH
A sequence of separate research claims	Independent implementation, nulls, replication, shadow mode	A single PASS implying clinical suitability

LIFENODE / TRAJECTORY MEDICINE / 12

Data governance, human factors and regulation

A clean research artifact can still be unfit for real-world clinical decisions.

12.1 Privacy and research governance

Use de-identified or appropriately governed records under their access conditions and licences. Track cohort definition, demographic balance, missingness, exclusion criteria, annotation provenance and rights to redistribute derived artifacts. Do not publish identifiable patient records or transfer protected health information into general-purpose public tools.

Any prospective involvement of participants needs the relevant ethics review or documented exemption and clear separation between routine care and the exploratory output. In silent shadow mode, a future model may be evaluated without delivering alerts to patients or altering treatment.

12.2 Human factors

Longitudinal baselines may adapt to individuals yet fail during non-wear, travel, work shifts, sleep disruption or medication change. A robust model needs explicit abstention and quality policies. Disparities may arise across age, sex, comorbidities, care settings and device brands; optical sensing introduces skin-tone and perfusion considerations.

The clinical burden of false positives must be assessed prospectively. A signal that creates anxiety, unnecessary visits or missed events cannot be considered a success solely on a ROC curve.

12.3 Regulatory trajectory

Future software analysing physiological signals to provide disease warnings may fall within medical-device oversight depending on intended use and jurisdiction. FDA clinical decision-support guidance and AI-enabled device materials [3,4] motivate a risk-based lifecycle, but this whitepaper does not perform a device classification or establish regulatory clearance.

The research-to-product ladder requires separate claims and evidence: research programme → validated analytical method → decision-support candidate → regulated medical product, where applicable. No figure in this edition skips those stages.

LIFENODE / TRAJECTORY MEDICINE / 13

Six-month research cycle / Nov 2026–Apr 2027

A constrained sequence with release criteria, not an open-ended promise of hardware or clinical deployment.

The six-month plan prioritises one end-to-end falsifiable workflow. Each month produces an inspectable artifact and an exit condition. The dates are a proposed work schedule, not a claim that experiments or replication have already been completed. A single researcher may need collaboration or compute resources for independent execution.

MONTH	PRIMARY WORK	DECISION ARTIFACT
NOV 2026	Freeze endpoint, dataset, splits, baseline hierarchy, metric estimator and null plan.	Protocol; data manifest; preregistration; numerical spec.
DEC 2026	Implement loaders, baseline features, synthetic tests and deterministic reconstruction.	Reproducible package; unit tests; initial baseline.
JAN 2027	Locked retrospective benchmark; subject-wise comparison; ablations and nulls.	Primary analysis and confidence intervals.
FEB 2027	Independent audit of resampling, windows, confounding and acquisition.	Sensitivity report; frozen candidate or rejection.
MAR 2027	Replication with locked model on an independent compatible cohort.	Replication/non-replication with no test retuning.
APR 2027	Release code, methods, manifests, hashes, results and a decision memo.	DOI-ready archive; continue/narrow/redesign/stop.

13.1 Independent HMF side stream

If resources allow, HMF may run in parallel as a materials research thread. Its deliverable is a bench report of transfer response, phase error, electrode comparisons, drift, controls and failure modes. It must not be used as evidence for a human clinical endpoint or silently inserted into the Zero-Build baseline experiment.

13.2 Explicit non-deliverables

- No patient-facing alert application or automated clinical triage.
- No active entrainment or closed-loop treatment engine.
- No expensive quantum sensing hardware absent a documented measurement gap.
- No 24–48-hour preclinical warning claim generated from short ECG records.

RESEARCH PLATE 05 / ROADMAP

Six stations / a complete falsifiable cycle

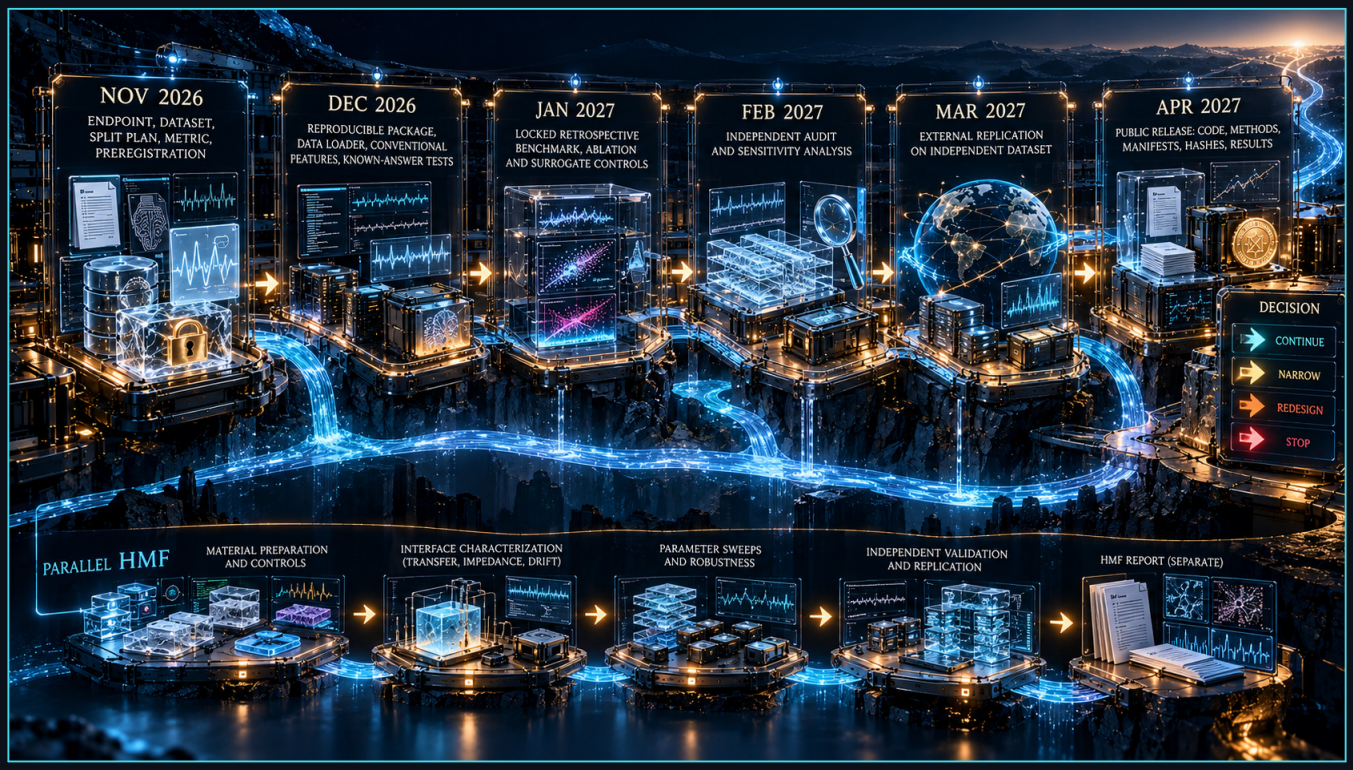


FIGURE 5. Six illustrated research stations from preregistration to DOI-ready public release, with a separate HMF engineering stream below the main track.

> HOW TO READ THIS PLATE
> Treat the six months as gated work packages. Independent audit and replication are not cosmetic checkpoints, and the parallel HMF lane is not evidence of patient readiness. Every station ends in an artifact and an explicit option to halt or revise.

EPISTEMIC STATUS Prospective schedule, not completed execution or validated outcomes.

REPRESENTS	REQUIRES A TEST	DOES NOT ESTABLISH
Versioned monthly research artifacts	Exit criteria, audits, locked external test	A guaranteed implementation or trial schedule

LIFENODE / TRAJECTORY MEDICINE / 14

Risks and portfolio decisions

The programme should know exactly what would force its central claim to shrink.

RISK	CONSEQUENCE	MITIGATION / DECISION
Metric ambiguity	θ changes meaning across documents or sampling conventions.	Frozen equation, units, estimator and normalisation.
Nonstationarity	Normal adaptation resembles abnormal trajectory drift.	Temporal validation, context variables and explicit abstention.
Leakage	Overlapping windows or patient identity inflates performance.	Subject/episode splits; independent audit.
Weak endpoint	“Phase drift” not tied to an adjudicated event.	Timestamped outcome and fixed clinical horizon.
Acquisition artefact	Contact, filtering or clock instability drive geometric features.	Raw-chain provenance and acquisition nulls.
Cross-scale overreach	Fungal or mitochondrial findings are treated as human clinical proof.	Separate mechanistic experiments by substrate.
Premature product	Research score is interpreted as medical advice.	Offline-only output, intended-use review and later regulatory plan.

14.1 Four defensible outcomes

CONTINUE. A frozen feature demonstrates reproducible incremental value, survives nulls and external replication, and merits a next prospective research protocol.

NARROW. An effect exists only in a defined modality, setting, endpoint or subgroup. The claim is constrained and retested there.

REDESIGN. The score is unstable or inadequately specified. Revise version, preregister new tests and preserve previous failures.

STOP. The relevant hypothesis or proposed warning use repeatedly fails the prespecified criteria, including acceptable alarm burden.

14.2 Do not confuse research persistence with claim preservation

A stop decision on the present clinical branch does not invalidate the entire LifeNode ecosystem. It says the tested specific use, dataset, model or score has not earned further clinical promotion. An open archive permits independent reinterpretation, new endpoints and alternative research directions without rewriting the historical record.

LIFENODE / TRAJECTORY MEDICINE / 15

Public release and provenance contract

What must remain after the research cycle, whatever the result?

A useful public release is more than a PDF. It identifies the exact question asked, materials analysed, what computations were run, what was held back, what failed and which conclusions were not licensed by the experiment. Provenance is the bridge between independent research contributors and future replication.

For each test, archive: study ID and amendment history; dataset version and licence; frozen split/subject identifiers where distributable; annotation rules; metric specification; code commit and environment lock; raw/derived hashes; preregistered statistics; null controls; uncertainty intervals; all outcomes including negative results; and reviewer/decision metadata.

15.1 Publication payload

- **METHODS** — human-readable protocol and one machine-readable analysis contract.
- **DATA LINKS** — accession identifiers and checksums, not prohibited redistribution of patient files.
- **CODE** — pinned commit, dependency lockfile, tests, instructions and reference outputs.
- **RESULTS** — passes, failures, invalids and inconclusives with common provenance schema.
- **DECISION** — explicit continue / narrow / redesign / stop and the single next falsifying experiment.

> DECISION MEMO RULE
> Release the negative and positive artifacts together. When evidence does not justify progression, the correct output is a reproducible limit - not silence and not a retroactively edited threshold.

RESEARCH PLATE 06 / RELEASE

Public archive / the open horizon



FIGURE 6. Imagined release dock containing code, methods, manifests, hashes, research records and a decision monolith with four admissible outcomes.

HOW TO READ THIS PLATE

The horizon stands for future questions, not clinical deployment. The vessel may depart with a negative result: intellectual progress here means an audit trail and a defensible next decision, not a predetermined success narrative.

EPISTEMIC STATUS Conceptual conclusion to the first proposed TMF research cycle; the depicted archive is not evidence of completed clinical validation.

REPRESENTS	REQUIRES A TEST	DOES NOT ESTABLISH
Methods, code, hashes, decisions	Reproducibility, licence compliance, negative outcomes	A released or authorised medical product

LIFENODE / TRAJECTORY MEDICINE / 16

Conclusion / measured ambition

Process > State is the organising research hypothesis. Measurements decide which part of it survives.

TMF can ask a serious question without overstating its answer: how much information about a physiological process is lost when analysis reduces a time series to conventional summary variables, and can that information be recovered in a robust, reproducible and useful way? The answer will depend on modality, endpoint, acquisition and the baseline being challenged.

SPRIZ is a specific biological structure–function example; slow fungal spikes are a distinct distributed electrophysiology example. Delay embedding is established mathematics under assumptions. Finsler structures, symplectic interpretations and NLSE analogies are not substitutes for evidence. ASCALON remains a project-defined metric, and HMF remains a separate material-instrument programme.

The near-term proof obligation is deliberately modest: one endpoint, one reproducible analytical implementation, strong comparators, strict partitioning, explicit nulls, an independent replication and a publication-ready failure record. A well-run falsification cycle can justify expansion, narrow an attractive idea or terminate a weak clinical proposal. All three can be intellectually productive.

The next legitimate stages are ordered: **research programme first; validated analytical method second; clinical decision-support candidate third; medical product only when evidence, safety and regulation permit.** A visual theory of process becomes science when it tells an independent researcher what would make it fail.

> LIFENODE / FINAL POSITION
> BUILD THE INSTRUMENT. FREEZE THE QUESTION. PRESERVE THE NEGATIVE. PUBLISH THE TRAJECTORY OF THE CLAIM.

LIFENODE / TRAJECTORY MEDICINE / APPENDIX A

Minimum experiment record

A reproducible claim requires a portable machine-readable chain from acquisition to decision.

FIELD	MINIMUM RECORD
Study identity	Protocol ID, version, date, authorship/operator, preregistration, amendments.
Endpoint	Target definition, event time, observation horizon, exclusions, censoring.
Dataset	Source accession/DOI, licence, versions, participant/record identifiers and hashes.
Acquisition	Device, firmware, clock, channels, ADC, filters, calibration and quality intervals.

FIELD	MINIMUM RECORD
Preprocessing	Full operation order, fitting partition, outlier and missing-data handling.
Embedding	τ/m selection method, training-only fitting, windows and sensitivity outputs.
Metric	Equation version, units, discrete estimator, normalisation and guards.
Comparators	Strong baseline hierarchy, model class, tuning budget and split plan.
Nulls	Time/label/phase surrogates, repetitions, seeds and expected negative outcomes.
Evaluation	Primary metric, uncertainty, calibration, subgroup and alarm-burden analyses.
Decision	PASS/FAIL/INVALID/INCONCLUSIVE, rationale, reviewer, next test.
Provenance	Code commit, environment, output hashes, archived logs and DOI reference.

Implementation and review checklist

Use as a pre-release checklist, not as a claim that a test has passed.

Before the hold-out is opened

- One question and endpoint are frozen with an admissible analysis horizon.
- Data integrity, patient/episode partitioning and licences are verified.
- Metric implementation matches its equation on known-answer inputs.
- All training-fit transforms and parameter searches are scoped to training data.
- Baseline hierarchy and tuning parity are declared.
- Structured nulls and statistical acceptance criteria are preregistered.

At independent audit

- A second implementation checks numerical identity where appropriate.
- The score remains interpretable under benign sampling and scaling changes.
- Confounding, leakage and negative controls are independently inspected.
- Confidence intervals and decision limits accompany every positive claim.
- Failures cannot be deleted or folded into a new test without versioning.

At public release

- A reviewer can reconstruct each result from manifests, code and parameter files.
- Data-sharing constraints are respected.

- All figure plates are labelled as conceptual where not measurement-derived.
- The decision memo identifies continue, narrow, redesign or stop.

LIFENODE / TRAJECTORY MEDICINE / APPENDIX C

Claim registry and future branches

Keep the conceptual programme open while making every test finite.

CLAIM	CURRENT CLASS	NEXT OBSERVATION NEEDED
Trajectory features add information to ECG labels.	Testable TMF hypothesis.	Strong held-out comparison + independent replication.
θ detects early disease drift.	Unvalidated hypothesis.	Frozen estimator and long-horizon prospective event dataset.
θ < 0.70 is a medical boundary.	Unsupported threshold proposal.	Normalisation, calibration, prospectively justified cut-off.
HMF provides a phase-useful interface.	Materials transduction hypothesis.	Transfer/impedance/phase and controlled null experiments.
Mitochondrial/fungal motifs transfer to humans.	Cross-domain speculation.	Explicit mechanistic bridge and substrate-specific validation.
Clinical alert system is useful.	Outside present scope.	Risk assessment, human factors, prospective utility and regulation.

LIFENODE / TRAJECTORY MEDICINE / REFERENCES

Source register

External observations and mathematical formalisms are distinguished from internal LifeNode working documents.

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ATTRIBUTION / ILLUSTRATIONS

All six supplied illustrations are conceptual research artwork. They communicate proposed instruments, abstractions or protocols; none is a scientific micrograph, measured HMF output, clinical validation result or completed 2027 milestone. The external references do not imply endorsement of LifeNode.